

§4.5

Independent Events

When knowing one event tells us nothing about another

Today's Goal

- Understand the concept of statistical independence
- Identify whether two events are independent or dependent
- Apply the multiplication rule for independent events
- Distinguish between independent events and mutually exclusive events
- Solve problems involving multiple independent events

What You Will Be Graded On

LT.7 Probability

I understand what the probability of an event represents and can apply basic rules of probability to calculate the probability of an event.

Drawing Two Marbles

A jar holds 5 red marbles and 5 blue marbles. You draw one marble, and it is red.

Does that change the chance that the next marble is red? What if you put the first marble back?

Independent and Dependent Events

Definition 4.5.1

Two events A and B are *independent* if the occurrence of one event does not affect the probability of the other event occurring. Mathematically, A and B are independent if $P(B \mid A) = P(B)$ and $P(A \mid B) = P(A)$.

Definition 4.5.2

Two events are *dependent* if the occurrence of one event changes the probability of the other event occurring.

Drawing Two Marbles

The jar holds 5 red and 5 blue marbles. The first marble drawn is red.

- Put back: the jar is 5 red out of 10 again. The chance the next marble is red stays $\frac{5}{10}$. The draws are independent.
- Not put back: 4 red remain out of 9. The chance drops to $\frac{4}{9}$. The draws are dependent.

Independent or Dependent?

Determine whether each pair of events is independent or dependent:

(a) Spinning a spinner twice: Event A = first spin lands on red, Event B = second spin lands on red.

Independent. The outcome of the first spin has no effect on the second spin.

(b) Rolling a die and flipping a coin: Event A = the die shows a 6, Event B = the coin shows heads.

Independent. The die has no effect on the coin. Each is a separate object.

Independent or Dependent?

Determine whether the pair of events is independent or dependent:

(c) Choosing two students without replacement from a class of 30 with 12 seniors: Event A = first student is a senior, Event B = second student is a senior.

Dependent. If the first student is a senior, there are only 11 seniors left among 29 students. So $P(B | A) = \frac{11}{29} \neq \frac{12}{30} = P(B)$.

Testing for Independence

We can test whether two events are independent using the following criterion.

Theorem 4.5.4 Test for Independence

Events A and B are independent if and only if

$$P(A \text{ and } B) = P(A) \cdot P(B).$$

If this equation does not hold, the events are dependent.

Using the Test for Independence

In a survey of 400 adults, 160 drink coffee every day (A), 100 have trouble sleeping (B), and 60 do both. Are the events A and B independent?

- $P(A) = \frac{160}{400} = 0.4$
- $P(B) = \frac{100}{400} = 0.25$
- $P(A \text{ and } B) = \frac{60}{400} = 0.15$

If A and B were independent, $P(A \text{ and } B)$ would equal $P(A) \cdot P(B) = 0.4 \times 0.25 = 0.10$.

Using the Test for Independence

400 adults: 160 drink coffee every day (A), 100 have trouble sleeping (B), and 60 do both.

$$P(A \text{ and } B) = 0.15 \text{ and } P(A) \cdot P(B) = 0.10.$$

Since $P(A \text{ and } B) = 0.15 \neq 0.10 = P(A) \cdot P(B)$, the events are dependent.

This makes sense: coffee and sleep are likely related. People who drink coffee every day tend to have more trouble sleeping.

Multiplication Rule (Independent)

When events are independent, the multiplication rule simplifies because $P(B \mid A) = P(B)$.

Theorem 4.5.6 Multiplication Rule for Independent Events

If A and B are independent events, then

$$P(A \text{ and } B) = P(A) \cdot P(B).$$

This rule extends to any number of independent events:

$$P(A_1 \text{ and } A_2 \text{ and } \cdots \text{ and } A_k) = P(A_1) \cdot P(A_2) \cdots P(A_k).$$

Three Spins of a Spinner

A spinner has 4 equal sections: red, blue, green, and yellow. It is spun three times. Find the probability that all three spins land on blue.

Each spin is independent, and $P(\text{blue}) = 0.25$ for each spin:

$$P(\text{all three blue}) = 0.25 \times 0.25 \times 0.25 = 0.25^3 = 0.015625.$$

There is about a 1.6% chance of three blues in a row.

Six Batteries

Phone batteries from a factory pass inspection 98% of the time. If 6 batteries are tested independently, what is the probability that all 6 pass?

Since each battery is tested independently:

$$P(\text{all 6 pass}) = 0.98^6 \approx 0.8858.$$

There is about an 88.6% chance that all 6 batteries pass.

“At Least One” Problems

A common type of problem asks for the probability of “at least one” success in multiple independent trials. The complement rule is especially useful here.

Procedure 4.5.9 “At Least One” Strategy

To find $P(\text{at least one success})$ in independent trials:

1. Find $P(\text{no successes}) = P(\text{all failures})$.
2. Use the complement: $P(\text{at least one success}) = 1 - P(\text{no successes})$.

At Least One Gold

A spinner has 5 equal sections, and one of them is gold. It is spun 3 times. What is the probability that at least one spin lands on gold?

The complement of “at least one gold” is “no gold” (all three spins are not gold).

For each spin, $P(\text{not gold}) = \frac{4}{5}$. Since the spins are independent:

$$P(\text{no gold}) = \left(\frac{4}{5}\right)^3 = \frac{64}{125} = 0.512.$$

$P(\text{at least one gold}) = 1 - 0.512 = 0.488$. There is a 48.8% chance.

Two Backup Generators

A hospital has two independent backup generators. Each generator has a 5% chance of failing during a power outage. What is the probability that at least one generator works during an outage?

The complement of “at least one works” is “both fail.”

$$P(\text{both fail}) = 0.05^2 = 0.0025.$$

$$P(\text{at least one works}) = 1 - 0.0025 = 0.9975.$$

There is a 99.75% chance that at least one generator works. This illustrates why redundant backup systems are so effective.

Now You Try: Smoke Detectors

You Try 4.5.3. A smoke detector has a 10% chance of failing to detect a fire. If a house has 3 independent smoke detectors, what is the probability that at least one detector will detect a fire? Round to 4 decimal places.

You Try 4.5.3: Checking Your Answer

Each detector fails 10% of the time, and there are 3 independent detectors.

$P(\text{fail}) = 0.10$ for each detector.

$$P(\text{all fail}) = 0.10^3 = 0.001$$

$$P(\text{at least one detects}) = 1 - 0.001 = 0.999$$

Independence vs. Mutually Exclusive

Mutually Exclusive Events

- Cannot occur at the same time.
- $P(A \text{ and } B) = 0$.
- Knowing A occurred tells us B did NOT occur.
- Mutually exclusive events are *dependent* (unless one has probability 0).

Independent Events

- Can occur at the same time.
- $P(A \text{ and } B) = P(A) \cdot P(B)$.
- Knowing A occurred tells us nothing about B .
- Independent events are NOT mutually exclusive (unless one has probability 0).

Mutually Exclusive Implies Dependent

Warning 4.5.13

Mutually exclusive events cannot be independent (unless at least one event has probability 0). This is because if you know one event occurred, you automatically know the other did not—so the events are dependent.

A Red Card and a Black Card

One card is drawn from a standard deck. Let A = drawing a red card and B = drawing a black card.

(a) Are A and B mutually exclusive?

Yes, they are mutually exclusive. A single card cannot be both red and black.

(b) Are A and B independent?

No, they are NOT independent. If we know that A occurred (the card is red), then we know for certain that B did NOT occur, so

$$P(B \mid A) = 0 \neq P(B) = 0.5.$$

Independent but Not Mutually Exclusive

One card is drawn from a standard deck.

Name two events that are independent and NOT mutually exclusive.

One answer: A = drawing a red card and B = drawing a face card.

Not mutually exclusive: the jack, queen, and king of hearts and of diamonds are both red and face cards.

Independent: $P(B \mid A) = \frac{6}{26} = \frac{3}{13}$, and $P(B) = \frac{12}{52} = \frac{3}{13}$, so $P(B \mid A) = P(B)$.

Classify Each Pair

Classify each pair of events as mutually exclusive, independent, both, or neither.

- (a) One roll of a die: rolling a 2, rolling a 5.

Mutually exclusive. One roll cannot show both.

- (b) Spin a spinner and draw a card: the spinner lands on red, the card is a king.

Independent. The spinner has no effect on the card.

Classify Each Pair

Classify each pair of events as mutually exclusive, independent, both, or neither.

(c) One card: drawing a heart, drawing a king.

Independent: $P(\text{heart and king}) = \frac{1}{52} = \frac{1}{4} \cdot \frac{1}{13}$. **Not mutually exclusive: the king of hearts.**

(d) One card: drawing a heart, drawing a red card.

Neither. Every heart is red, so both can happen. $P(\text{heart and red}) = \frac{1}{4}$,
but $\frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$.

Summary of Key Formulas

Concept	Formula
Complement	$P(E') = 1 - P(E)$
Addition Rule (Mutually Exclusive)	$P(A \text{ or } B) = P(A) + P(B)$
General Addition Rule	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
Conditional Probability	$P(B A) = \frac{P(A \text{ and } B)}{P(A)}$
General Multiplication Rule	$P(A \text{ and } B) = P(A) \cdot P(B A)$
Multiplication Rule (Independent)	$P(A \text{ and } B) = P(A) \cdot P(B)$
Test for Independence	Events are independent if $P(A \text{ and } B) = P(A) \cdot P(B)$

What to Take With You

- Independent: knowing one event occurred tells us nothing about the other.
- To test, compare $P(A \text{ and } B)$ with $P(A) \cdot P(B)$.
- For independent events, multiply the probabilities.
- “At least one” is $1 - P(\text{none})$.
- Mutually exclusive events with nonzero probabilities are always dependent.